

Commands and lifecycle

Siddharth M. Bhatia · siddharth@sidmb.com

August 8, 2026

Contents

Launch and attach	1
Durable tasks	2
Inspect sessions	2
Synchronize and compare	3
Continuous synchronization	3
Forward local ports	5
Manage configured targets	5
Stop, remove, and uninstall	6

Launch and attach

```
fwd                                # human reuse workflow: attach or
    create interactively
fwd up                             # launch the configured default;
    stay local
fwd up runpod codex                # positional target and coding
    agent
fwd up --target work --agent codex
fwd up --new codex                # create a second session for
    this project
fwd up -- python train.py --epochs 10
fwd attach SESSION                # human terminal only
```

`fwd up` is idempotent and also repairs partially prepared sessions. Use `--new` only when you want a separate remote resource. Agent launches attach automatically only in a human terminal; `--detach` always stays local.

An explicit command runs as a durable task and returns its exit status. After two seconds, `Ctrl-C` cancels that task and `Ctrl-B` backgrounds the local viewer. `fwd attach` enters the primary tmux session; detach with `Ctrl-B D`.

Durable tasks

`fwd send` runs work in an existing session and never starts or restarts compute:

```
fwd send -- pytest -q
fwd send --name SESSION --detach -- python train.py
fwd send --ls --json
fwd send TASK_ID          # replay the log and continue
    following
fwd send TASK_ID --stop    # cancel only this task
fwd send cancel           # cancel all queued tasks
fwd send cancel all       # cancel all active tasks
```

Use `--` before commands whose flags belong to the remote program. Invoke a shell explicitly for pipes, redirects, and globs:

```
fwd send -- bash -lc 'cat outputs/*.json | jq .'
```

On supported backends, `--stop-after` queues remote-owned shutdown after new work completes. `fwd send stopafter` queues shutdown after all current work, and `fwd send cancel stopafter` disarms it. Lambda does not support remote stop-after; use local `fwd stop` after retrieving results.

Inspect sessions

```
fwd ls
fwd ls --all-projects
fwd ls --json
fwd ls --columns backend,status,ports
fwd doctor --json
fwd info --json
```

Rich tables are used in terminals. Redirected or agent output defaults to Markdown; `--json` provides stable machine-readable output. Progress and diagnostics stay on stderr.

Commands accept exact session names and, where unambiguous, target or backend aliases. Use selectors positionally with `attach`, `stop`, `rm`, and `diff`; use `--name` with `send`, `push`, and `pull`. `stop` and `rm` accept multiple selectors.

Synchronize and compare

```
fwd diff                    # unified diff; no changes to
    either side
fwd diff -q                # exit 0 same, 1 different, 2 error
fwd push                   # mirror local files to remote
fwd pull                   # additive whole-project download
fwd pull outputs/ logs/   # additive path-scoped download
```

Synchronization honors `.gitignore`, `.fwdignore`, and configured exclusions. Upload includes `.git/` so remote agents retain repository context; pull and diff exclude `.git/`, so pulling does not transfer remote commits. Export a patch or Git bundle when you need a remote commit locally.

Push mirrors the local synchronized tree and normally deletes remote-only synchronized files. Pull is always additive and never deletes local files. Uploads are capped by `sync.max_size_gb` (1 GB by default) and use a staging directory so an interrupted or over-limit upload does not replace the live remote project.

Continuous synchronization

`fwd sync` keeps the local project and the remote directory converged in both directions while a session runs, instead of waiting for an explicit push or pull. It is built on [Mutagen](#), is **off by default**, and is stored per target:

```
fwd sync on                # enable for this target and start
    it now
fwd sync status            # configured intent, live state,
    and conflicts
fwd sync status --json
fwd sync off               # stop syncing; the remote
    directory is left as-is
```

```
fwd sync on SESSION          # act on a named session instead of
    this directory's
```

Toggling takes effect immediately, including on a session that is already running. Enabling it also enables it for future launches on that target; `fwd up` and `fwd attach` start the sync automatically whenever it is on.

Key properties:

- **.git is never continuously synced.** Both sides run Git concurrently, and a half-propagated index or packfile corrupts the repository. `fwd push` and `fwd pull` still carry `.git`, and remain the way to move repository state.
- **Conflicts are never resolved destructively.** The mode is Mutagen's two-way-safe, so a file edited on both sides is reported by `fwd sync` status rather than overwritten. Settle one by deleting the copy you do not want on one side; the surviving version then propagates. Editing the losing side again does not help — that is simply another competing change.
- **The initial transfer still goes through rsync.** `fwd up` performs its usual size-bounded push, and Mutagen only moves steady-state deltas afterwards, so `sync.max_size_gb` still governs the large transfer.
- **Ignore rules are captured when the sync starts,** translated from `.gitignore` (including nested files), `.fwdignore`, and `sync.exclude`. After editing those, run `fwd sync off && fwd sync on` to pick up the change.
- **Not every transport supports it.** Mutagen installs a remote agent over `scp`, which the RunPod SSH proxy cannot run; `fwd` warns and skips continuous sync there.

`fwd` installs and drives Mutagen itself: the remote agent installs automatically on first connection, and the first local use offers to run `brew install mutagen-io/mutagen/mutagen` when Homebrew is available. `fwd` runs its own isolated Mutagen daemon under `~/ .fwd/mutagen`, so it never disturbs a Mutagen daemon you run yourself.

Forward local ports

```
fwd ports 3000                # local 3000 to remote 3000
fwd ports work 8080:3000      # local 8080 to remote 3000
fwd ports --ls
fwd ports --close 3000
fwd ports work --close        # close all forwards for one session
fwd up --ports 3000 --ports 8080:3000
```

Forwards bind local and remote loopback only. fwd refuses occupied local ports before changing tunnel state and tracks persistent OpenSSH control masters across CLI processes.

Manage configured targets

fwd targets manages saved targets — the [targets.NAME] configuration entries that describe *how to reach* compute. The session commands above manage the compute itself.

```
fwd targets ls                # name, backend, key connection
    detail, default marker
fwd targets ls pod            # only targets whose name contains
    "pod"
fwd targets add ssh --host my-box --target-name work
fwd targets info work         # resolved values, which are
    defaults, and tracked sessions
fwd targets update work       # setup again, prefilled with the
    current values
fwd targets rm work           # remove the configuration entry
    only
```

fwd setup remains a permanent alias for fwd targets add. All of these read local configuration and session state only; no provider is contacted. update and rm open a picker when no target is named and fail with an actionable message when nothing can answer a prompt; rm requires --force non-interactively. Removing a target never stops or destroys compute — use fwd rm for that — and removing the default repoints default_target at the sole survivor or clears it.

Stop, remove, and uninstall

```
fwd stop SESSION           # stop compute; retain configured
                             persistent storage
fwd rm SESSION              # destroy remote resources and
                             forget the session
fwd rm one two              # operate on several resolved
                             sessions
fwd rm --all                # destroy every tracked session
                             after confirmation
fwd uninstall               # remove local fwd data, skill,
                             and completion
```

`stop` and `rm` check the remote Git worktree and refuse when reachable work is dirty. Resolve that work with `fwd diff`, `fwd pull`, or a remote commit before retrying. Force flags explicitly accept possible data loss.

`fwd rm` is irreversible. `fwd uninstall` does not destroy remote resources and should normally follow `fwd rm --all`.

See [Troubleshooting](#) for recovery and safety details.